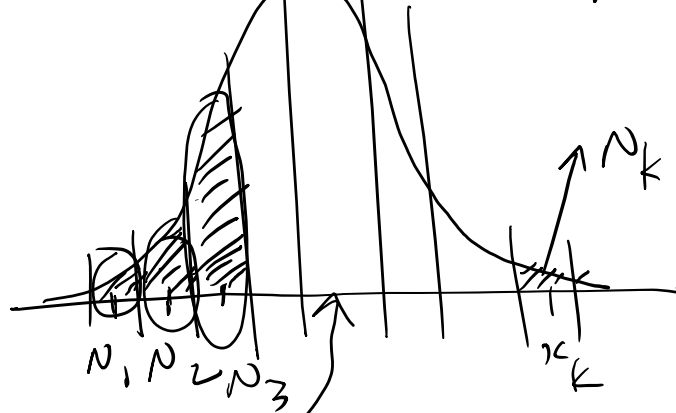
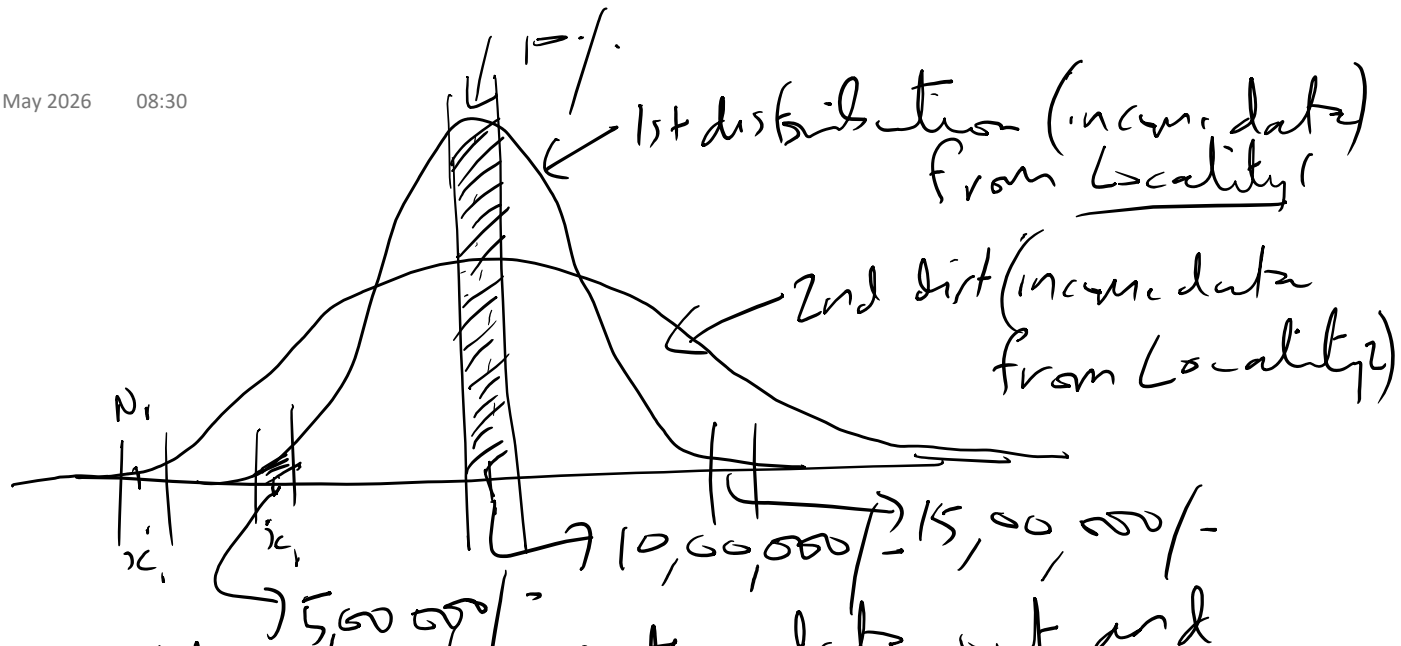


Mean, median, mode $\rightarrow$ not informative enough

same mean, median, mode
 $\rightarrow$ very different distributions



$$\text{Mean} = \frac{N_1 x_1 + N_2 x_2 + N_3 x_3 + \dots + N_k x_k}{(N_1 + N_2 + \dots + N_k)}$$



We can spread the data out and still have the same mean

$(X_1 - \mu)$ = deviation of score 1 from mean

$(X_2 - \mu)$ = deviation of score 2 from mean

$$\frac{(X_1 - \mu) + (X_2 - \mu) + \dots + (X_N - \mu)}{N} = \text{Average deviation}$$

$$\frac{(X_1 + X_2 + \dots + X_N) - N\mu}{N} = \frac{N\mu - N\mu}{N} = 0$$

$$\frac{X_1 + X_2 + \dots + X_N}{N} = \mu \quad \left[\text{By definition} \right]$$

$\uparrow$
 Mean

One idea we have

$$\frac{|X_1 - \mu| + |X_2 - \mu| + \dots + |X_N - \mu|}{N}$$

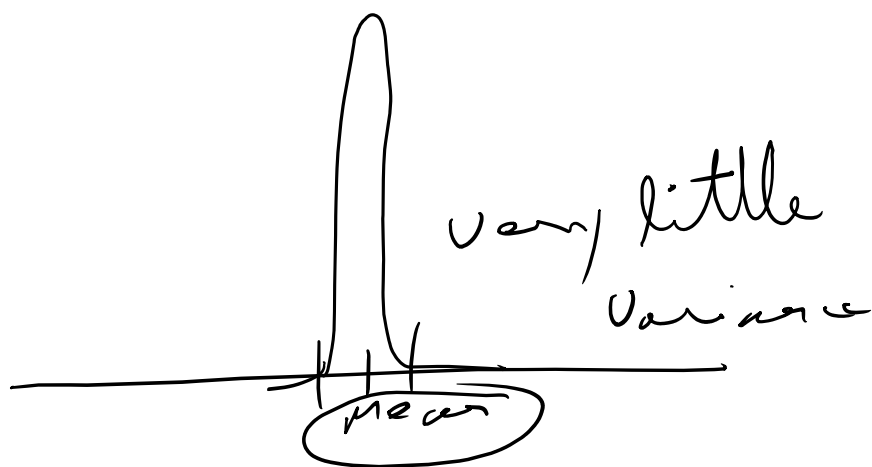
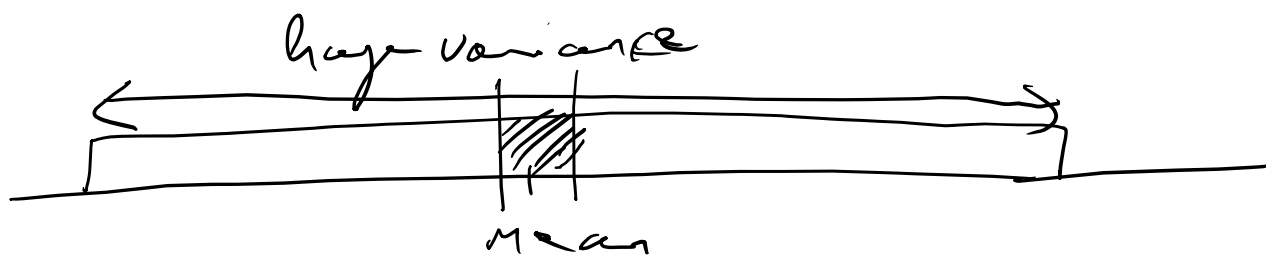
= average deviation [okay, but

there are some problems with this]

So take instead

$$\frac{(X_1 - \mu)^2 + (X_2 - \mu)^2 + \dots + (X_N - \mu)^2}{N}$$

= average squared deviation
= variance



Population mean

the country's population has a

Mean income

- in order to actually calculate
this mean, we need to look at every
single income [not possible]

We draw a sample of incomes from
the population

↳ say N samples

$$\text{Sample mean} = \frac{X_1 + X_2 + \dots + X_N}{N} = \bar{X}$$

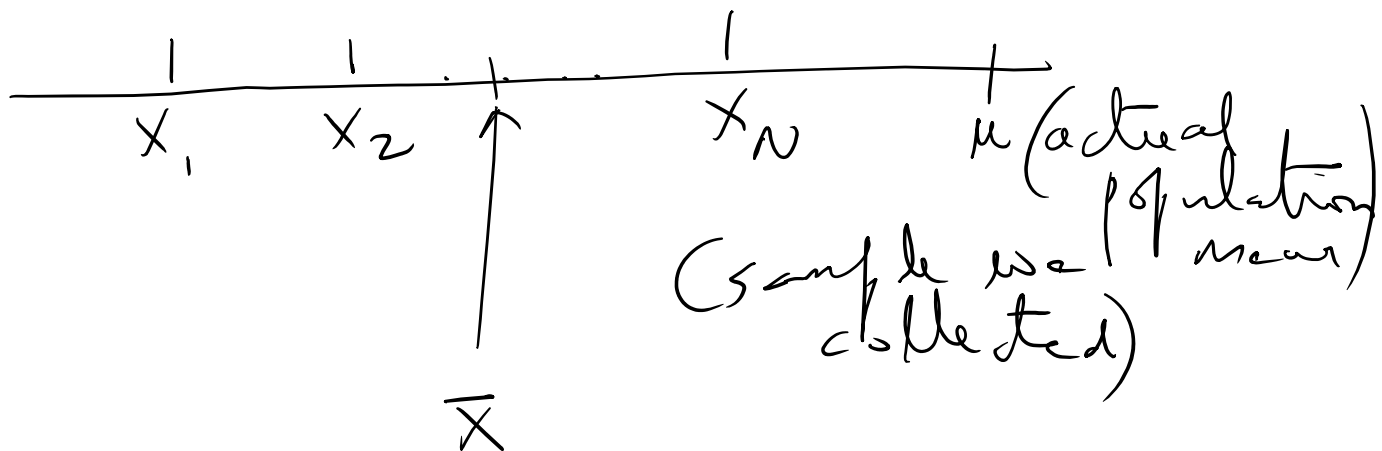
We pretend the sample mean is
equal to the population mean

A possible measure of

$$\text{Variance} = \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_N - \bar{x})^2}{N}$$

(natural formula)

sample has less variability than the whole population



$$\frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_N - \bar{x})^2}{N}$$

might not be all that high

$(x_1 - \bar{x})^2 + \dots + (x_N - \bar{x})^2$ will generally
be less than

$(x_1 - \mu)^2 + \dots + (x_N - \mu)^2$ where
 $\mu = \text{population mean}$ and $\bar{x}$ is
 sample mean

It turns out that we can compensate
 for this in a mathematically correct way
 by dividing the sum $(x_1 - \bar{x})^2 + \dots + (x_N - \bar{x})^2$
 by $(N-1)$, not N

$$\sqrt{\frac{\sum (X_i - \mu)^2}{N}}$$

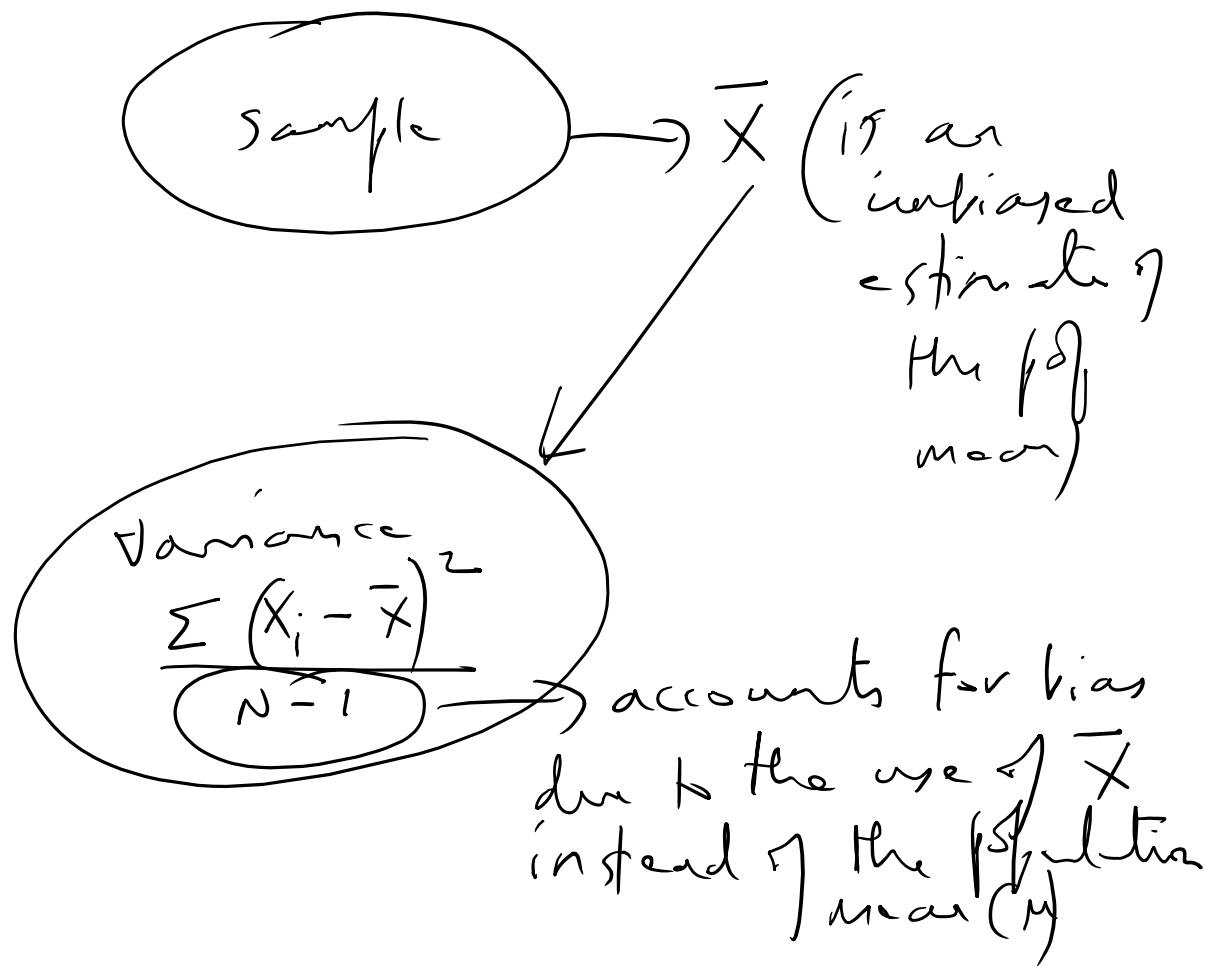
vs

population mean
average squared deviation

$$\sqrt{\frac{\sum (X_i - \mu)^2}{N}}$$

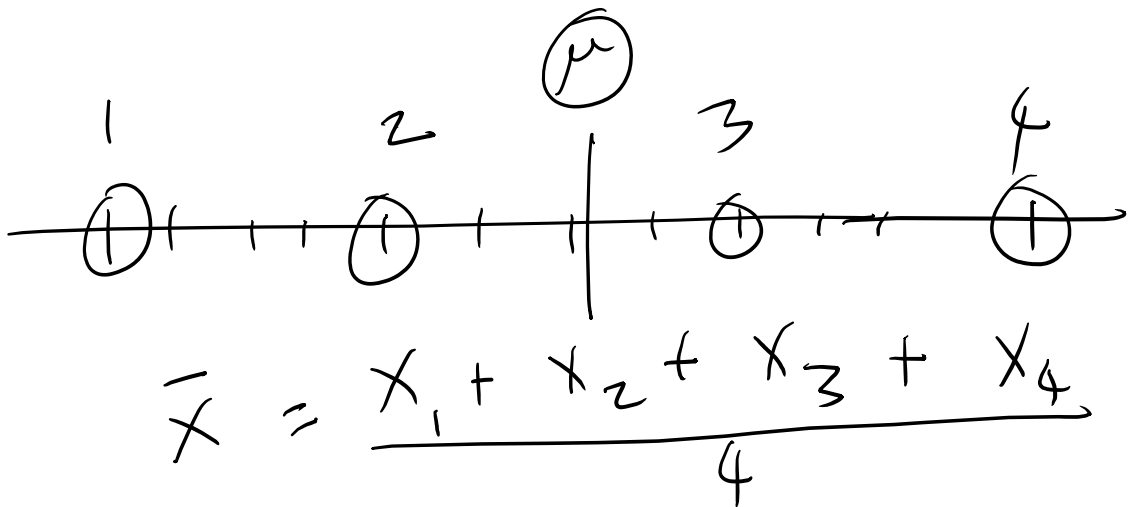
$$= \sqrt{\frac{\sum (X_i - \mu)^2}{N^2}}$$

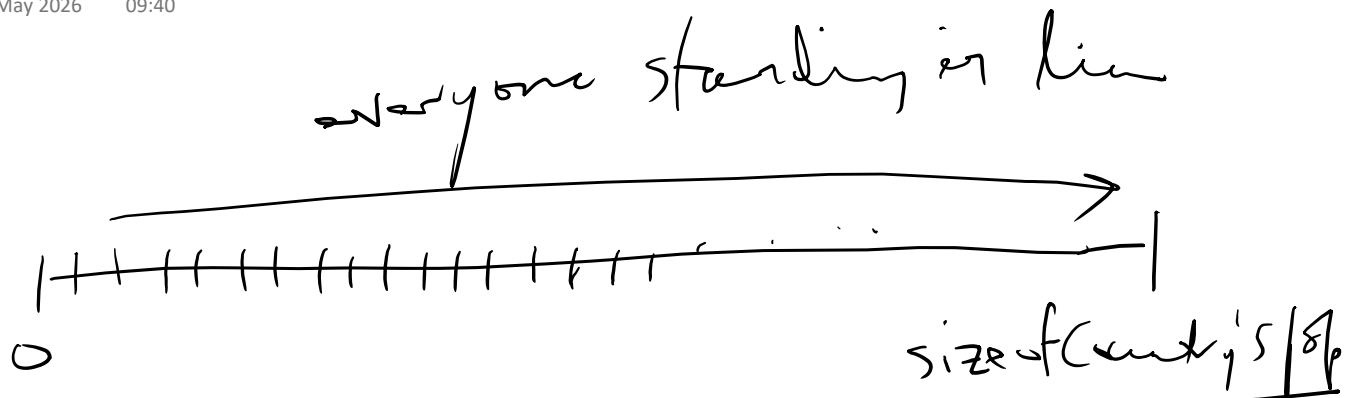
what meaning does this have?



It is not that $\bar{X} = \mu$, it is

that $E(\bar{X}) = \mu$ (no bias in $\bar{X}$)





generate a uniform random number
between 0 and size of County's pop
and pick that person from the county's
list and add to that sample.

Create a sample of N people

Compute $\bar{X} = \frac{X_1 + \dots + X_N}{N}$

Repeat the experiment $\rightarrow$ a new sample of N people

$\bar{X}' = \frac{X'_1 + \dots + X'_N}{N}$

Each $\bar{X}$ need not be μ

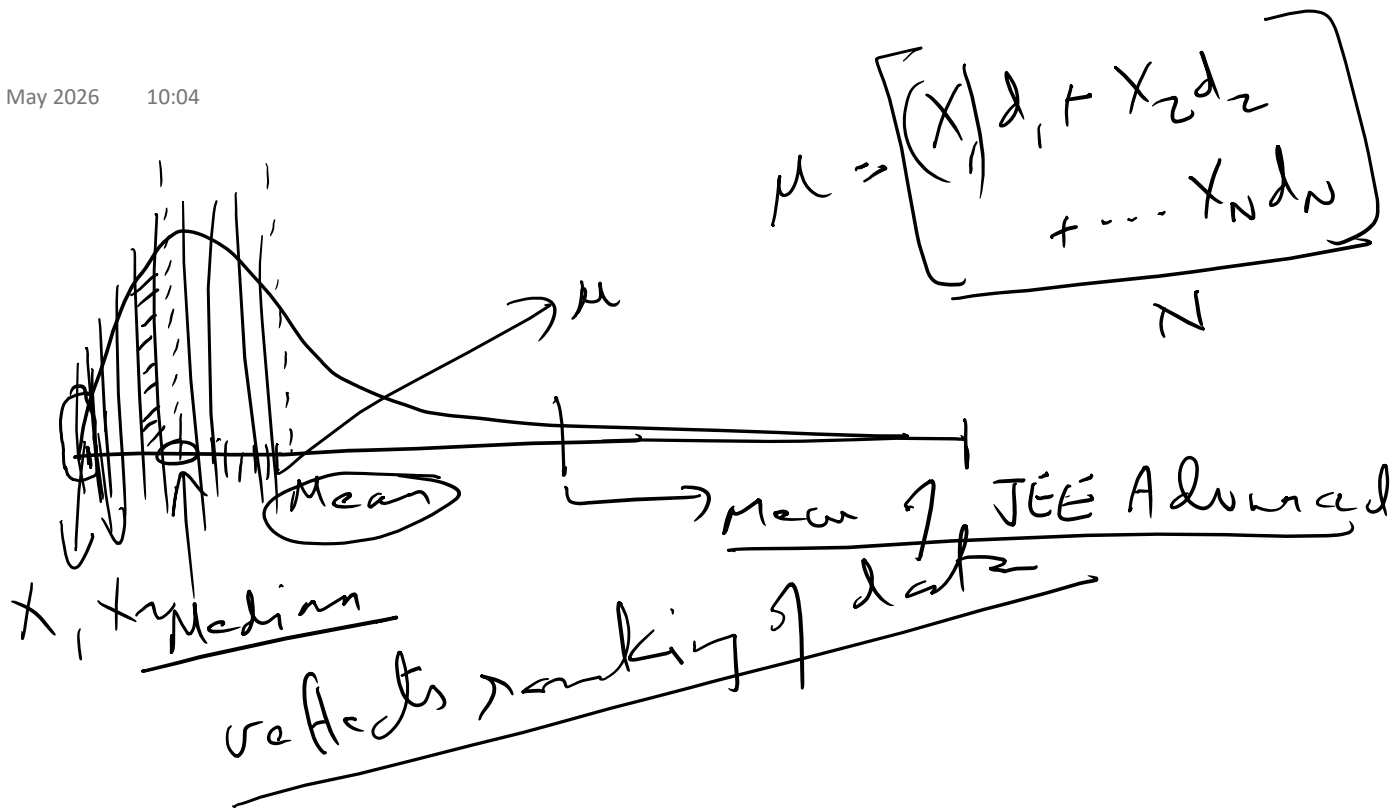
$$\text{but } E(\bar{X}) = \mu \quad [\text{unbiased}]$$

Average income of the N people
in the sample

$$x_1, x_2, \dots, x_N \rightarrow \bar{X} \quad [\text{one sample}]$$

$$x'_1, x'_2, \dots, x'_N \rightarrow \bar{X}' \quad [\text{another sample}]$$

$$x''_1, x''_2, \dots, x''_N \rightarrow \bar{X}'' \quad [\text{yet another sample}]$$



$$x_1, x_2, \dots, x_N$$

$$\text{Variance} = \frac{X_1(d_1 - \mu)^2 + X_2(d_2 - \mu)^2 + \dots + X_N(d_N - \mu)^2}{N}$$

"randomness" of an event is a
cover for a lack of knowledge
about the event

if we know the physics of tossing a
coin we could in principle
calculate the outcome
[it would then be deterministic]

Die $\rightarrow$ all possible outcomes
 $S = \{1, 2, 3, 4, 5, 6\}$

Event $= E = \{ \text{face of die is an odd number} \}$

$$E \subseteq S$$

$$E = \{1, 3, 5\}$$

Classical probability

$$P(E) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$$

$$= \frac{3}{6} = \frac{|\{1, 3, 5\}|}{|\{1, 2, 3, 4, 5, 6\}|}$$